



Xiaomi Tablet 4 Level 3 Maintenance Guide V01

Technical support internal document control: TSIMMPD9 Xiaomi Tablet 4 Level 3 Maintenance Guidance V01

Scope of application:

Analysis center, mainboard and whole machine maintenance factory

Change history:

First edition 2018-11-12

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1. Introduction of basic information

1.1 Product Overview

Product Overview:

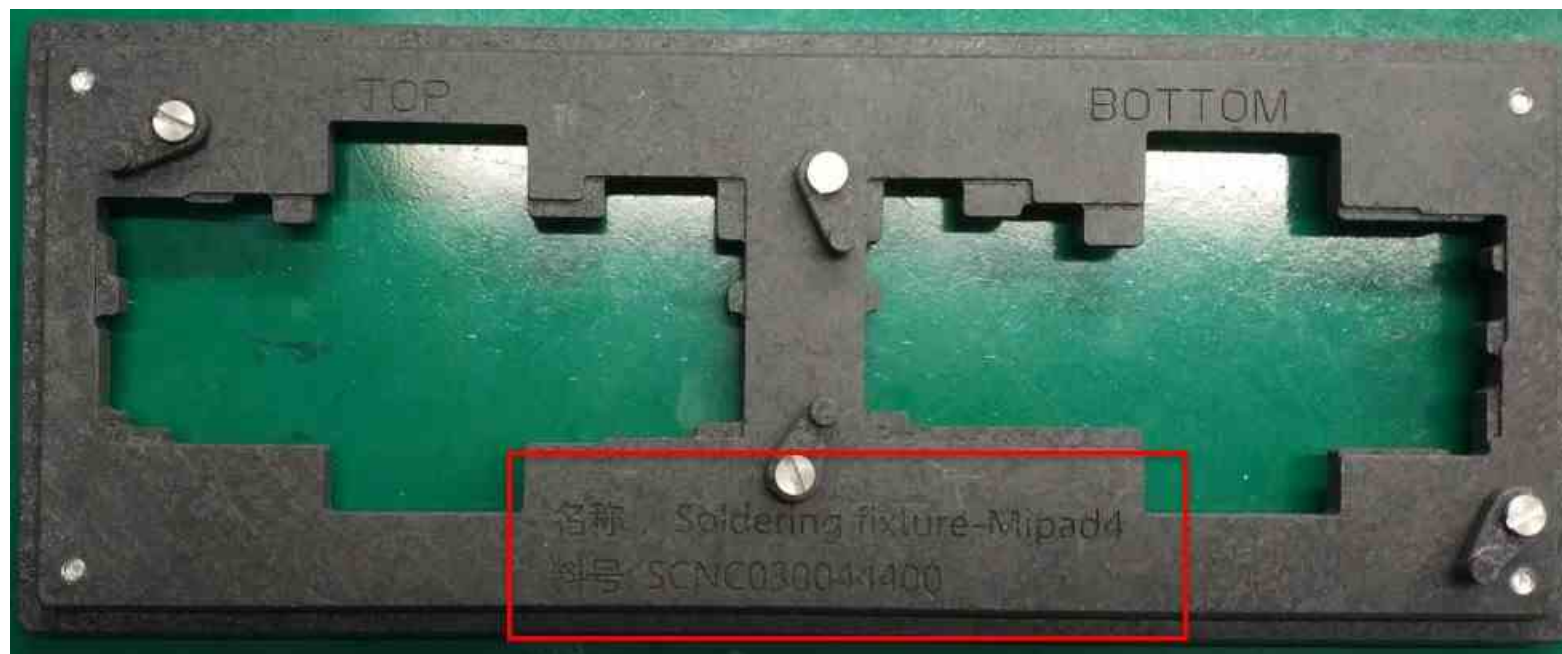
外观尺寸



处理器与容量	骁龙 660 处理器，最高主频 2.2GHz，Adreno 512 图形处理器，最高主频 650MHz，3GB+32GB / 4GB+64GB 两版本可选，LPDDR4x 双通道内存，eMMC 5.1 高速闪存，支持扩容256GB TF卡
重量与尺寸	屏幕：5.99英寸，高度：158.5mm，宽度：75.45mm，厚度：8.05mm，重量：180 g
续航与充电	6000mAh (typ) 大电量，支持5V / 2A充电，标配 5V / 2A 充电器
屏幕	8英寸，16:10 屏幕比例，1920x1200 FHD，283 ppi，1200:1对比度，支持护眼模式、标准模式、增强模式、夜光屏、智能适应模式下无级色温
拍照与摄像	1300万像素后置相机，f/2.0光圈，支持 HDR，支持美颜，支持 1080p 视频拍摄，1920x1080 30fps；500万像素前置相机，f/2.0光圈，支持前置美颜，支持 HDR 支持人脸识别
网络	WiFi 版：支持协议 802.11a/b/g/n/ac，支持 2.4G WiFi / 支持 5G WiFi / 支持 WiFi Direct，支持2×2Wi-Fi MIMO，支持 蓝牙5.0；WiFi + 4G LTE版：支持单卡不限运营商，单Nano-SIM卡槽，支持移动/联通/电信 4G，支持移动/联通4G+；支持频段：TDD-LTE:B34/B38/39/40/41(100M)*；FDD-LTE:B1/3/5/7/8；无线网络和WiFi版相同
无线网络	支持协议 802.11a/b/g/n/ac，支持 2.4G WiFi / 支持 5G WiFi / 支持 WiFi Direct，支持2×2Wi-Fi MIMO，支持 蓝牙5.0
导航与定位	WiFi 版：不支持导航服务； 4G LTE版：GPS·AGPS·GLONASS·Beidou
传感器	WiFi 版：陀螺仪、加速感应器、环境光传感器、霍尔传感器 4G LTE 版：陀螺仪、加速感应器、环境光传感器、霍尔传感器、电子罗盘
多媒体播放	音频格式：AAC LC、HE-AACv1 (ACC+)、HE-AACv2 (ACC+)、ACC、AMR-NB、AMR-WB、FLAC、MP3、PCM/WAVE、Vorbis 视频格式：H.263、H.264 AVC Baseline Profile (BP)、H.264 AVC Main Profile (MP)、H.264 HEVC、MPEG-4 SP、XVID、VP8、VP9、ASF/WMV、3G2、AVI、TS、MKV/WEBM、M4V、3GP/MP4、FLV

1.2 Special welding fixture for millet plate 4

Material code: SCNC030044400



1.3 Xiaomi flat plate 4 power supply adapter

Material code: SCNC030043000



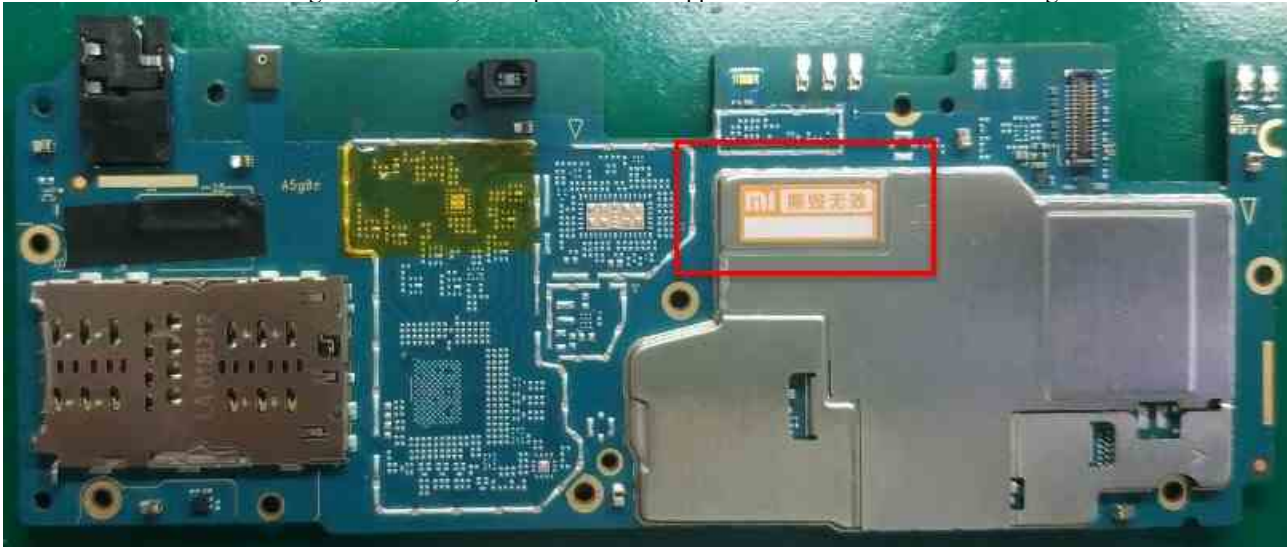
1.4 Millet flat plate 4 RF test jig

Material code: SCNC030044500

1.5 Pasting position and specification of maintenance label

Xiaomi Tablet 4
Maintenance Label: The pasting position is as shown in the figure:

The service label is 12mm long * 5mm wide, and is pasted on the upper left corner of SH2104 shielding frame.



1.6 Matters needing attention in motherboard maintenance

Note:

1. There is a binding relationship between CPU and EMMC. EMMC must be replaced at the same time when replacing CPU, and EMMC can be replaced separately. However, after replacing EMMC, it can be turned on normally in factory mode. If TEEKEY is not uploaded, it will be turned on and restarted in user mode.
2. When maintaining and heating the motherboard, it is necessary to take off the shielding frame of CPU shielding shell, so as to avoid other faults caused by the expansion and extrusion of heat-conducting silica gel after overheating.
3. Take anti-welding measures when welding the components near the key interface, display interface and earphone interface. These plastic components are easy to weld.
4. Enter CIT mode in user mode, click "Settings-All Parameters-Kernel Version", and click "Kernel Version" four times in succession to enter CIT test mode.

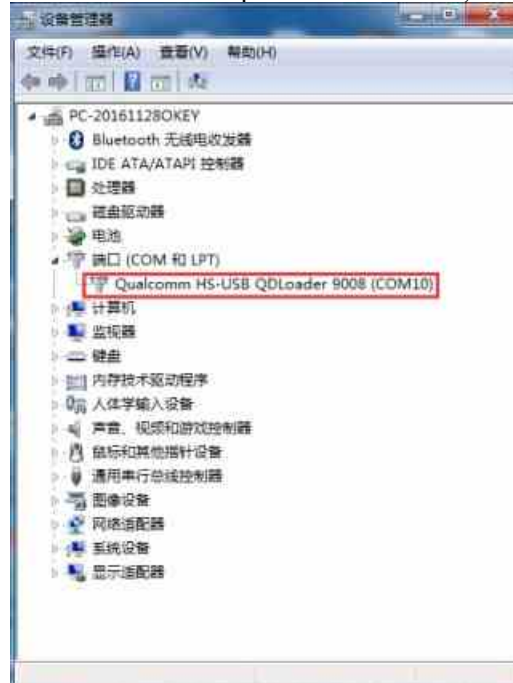
1.7 Brush mode

Brush platform: MiFlash

Special factory package for renaming (keep NV package): millet flat 4_D9_keeper NV factory package. zip
special factory package for maintenance (remove NV package): millet flat 4_D9_eraseNV factory package. zip

Brushing method: deep brushing

1. When the mobile phone is turned off, short-circuit TP2103 and TP2104 to make the mobile phone enter deep brush mode (identify port 9008).



Click "Delete All Options" at the bottom of MiFlash, click "Load Device", and click "Brush Machine".



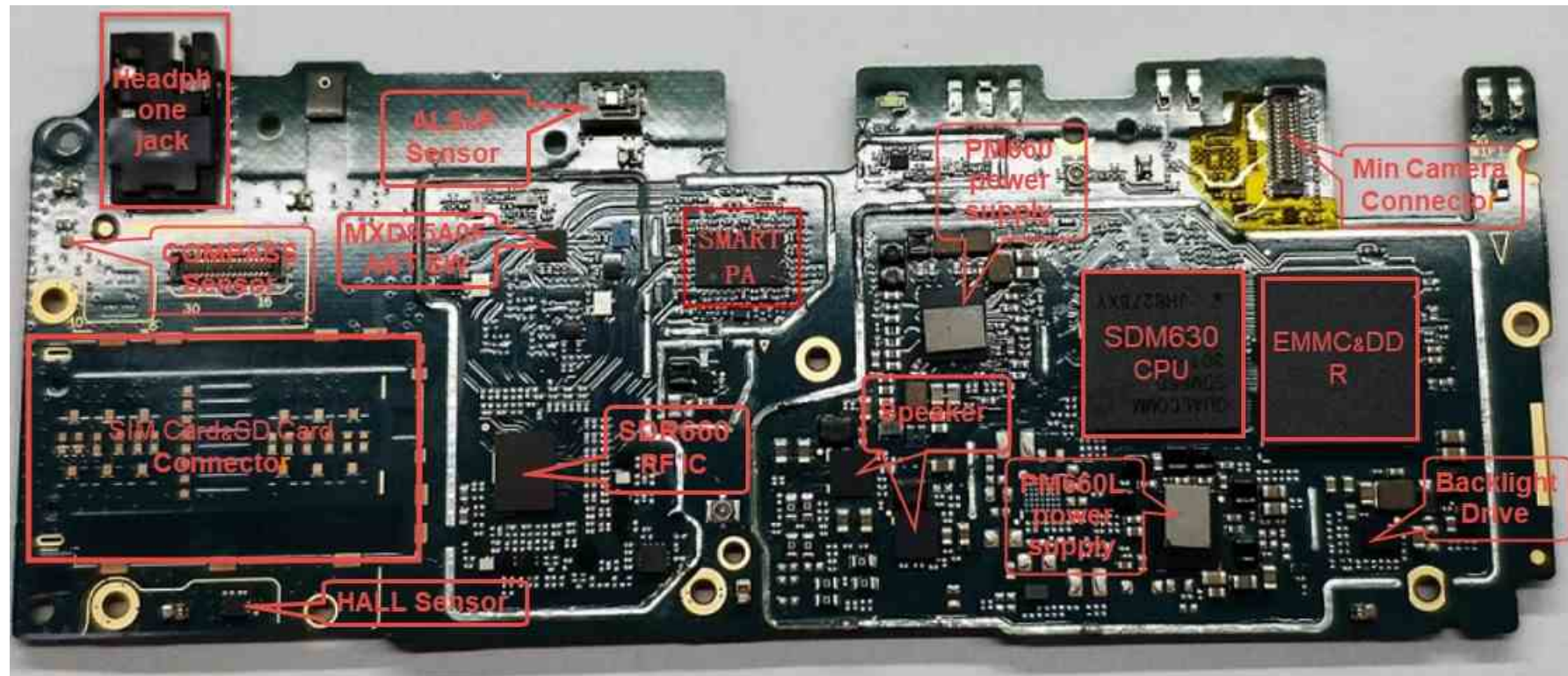
1.8 RF calibration test correlation

1. Deep brush factory software, write FSN with DT tool.
2. Open the calibration software, and connect the main receiving antenna and the auxiliary receiving two RF lines when the mobile phone is turned on. Connect the USB cable and the RF calibration of the mobile phone starts (refer to the calibration operation instruction (XTT) of D9_LTE version for details).
3. After successful calibration, use the "NV Read & Write V2.2 Tool" to write the flag bit (refer to the NV Read & Write V2.2 Operation Instructions for details).
4. After restarting with DT tool, you can directly insert the card to test the signal.

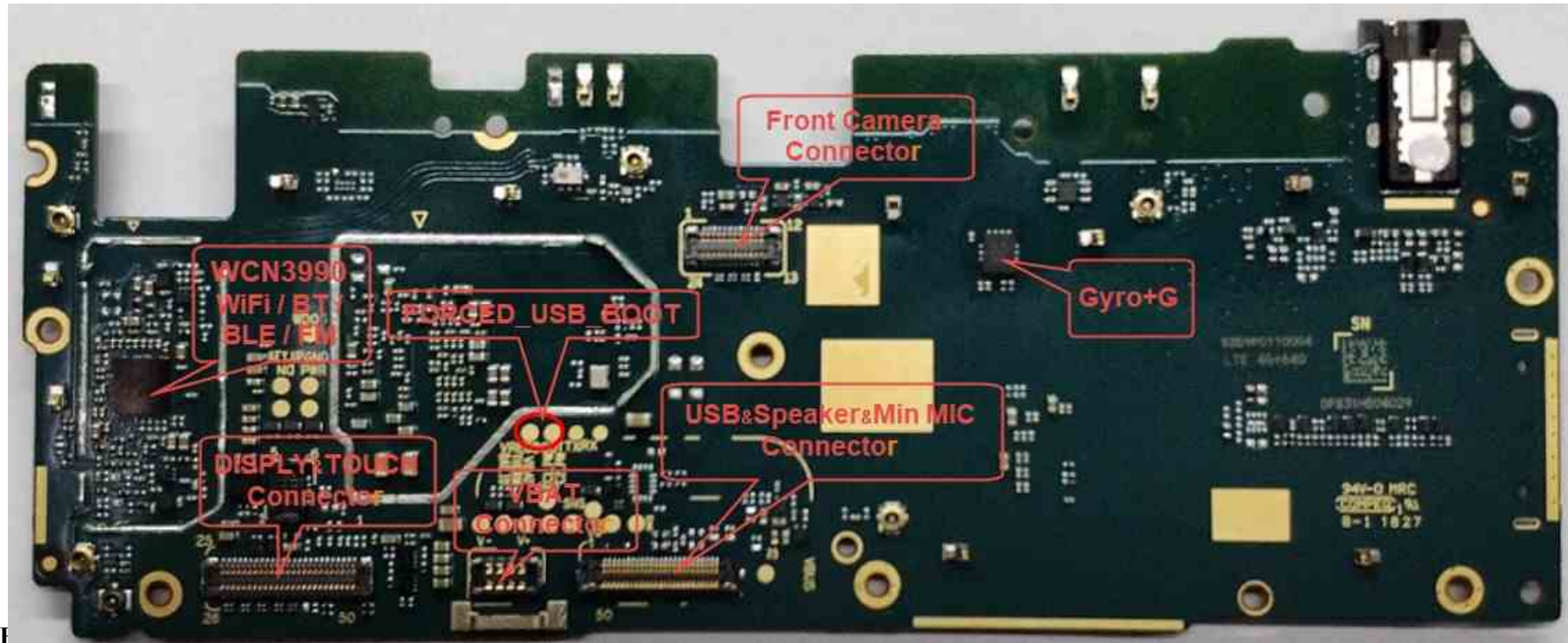
2. Brief introduction of motherboard module

2.1 Component Distribution Map of Xiaomi Tablet 4 Main Board

- TOP plane



- BOT surface



2.2 I

Boot Timing:

开机时序测量表		
Symbol	测量值	测量点
KYPD_PWR_N	1.8V	TP2805
VREG_BOB	3.3V	C1403
VREG_S5B_0P915	0.915V	L1705
VREG_S3B_S4B_0P87	0.87V	L1703
VREG_S2B_1P05	1.05V	L1702
VREG_L3A_1P0_SDR_DVDD	1V	C1406
VREG_S4A_2P04	2.04V	C1806
VREG_S5A_1P35	1.35V	C1313
VREG_L13A_1P8	1.8V	C1415
VREG_L10A_1P8	1.8V	C1412
VREG_L9A_1P8	1.8V	C1411
VREG_L6A_1P3	1.3V	C1408
VREG_L11A_1P8	1.8V	C1413
REF_MSM_HVPAD	1.25V	C1042
VREG_L10B_0P915	0.915V	C1717
VREG_L9B_0P87	0.87V	C1716
VREG_L1B_0P925	0.925V	C1034
VREG_L7B_3P125	3.125V	C1022
VREG_L2B_SDC2	2.95V	C1011
VREG_L5B_2P95	2.95V	C1712
VREG_S1B_1P125	1.125V	L1701
VREG_L4B_2P95	1.95V	C1711
VREG_L8A_1P8	1.8V	C2008
SLEEP_CLK	32KHz	R1213
VREG_S1A_0P87	0.87V	L1301
VREG_L14A_1P8	1.8V	C1416
VREG_L1A_1P225	1.225V	C1404
LNBBCLK	19.2MHz	R1201
PON_RESET_N	1.8V	R509
SDM_PS_HOLD	1.8V	TP2113

CPU (U2001) power supply signal and measurement points (yellow signals are also signals in startup timing):

CPU供电测量表		
Symbol	测量值	测量点
VREG_S5B_0P915	0.915V	L1705
VREG_L10B_0P915	0.915V	C1717
VREG_L10A_1P8	1.8V	C1412
REF_MSM_HVPAD	1.25V	C1042
VREG_L2B_SDC2	2.95V	C1011
VREG_S1A_0P87	0.87V	L1301
VREG_S3B_S4B_0P87	0.87V	L1703
VREG_L9B_0P87	0.87V	C1716
VREG_L1B_0P925	0.925V	C1034
VREG_L1A_1P225	1.225V	C1404
VREG_L7B_3P125	3.125V	C1022
VREG_L13A_1P8	1.8V	C1415
VREG_L15A_UIM1	C1021 (SIM1插卡)	2.9V
VREG_L17A_UIM2	C1026 (SIM2插卡)	2.9V
VREG_L6A_1P3	1.3V	C1023
VREG_L9A_1P8	1.8V	C1030
VREG_S6A	0.7V	C901
VREG_L5A_0P848	0.848V	C1032
VREG_S1B_1P125	1.125V	C1010
VREG_S2A_S3A_0P87	0.87V	L1302

U1201 chip bitmap:

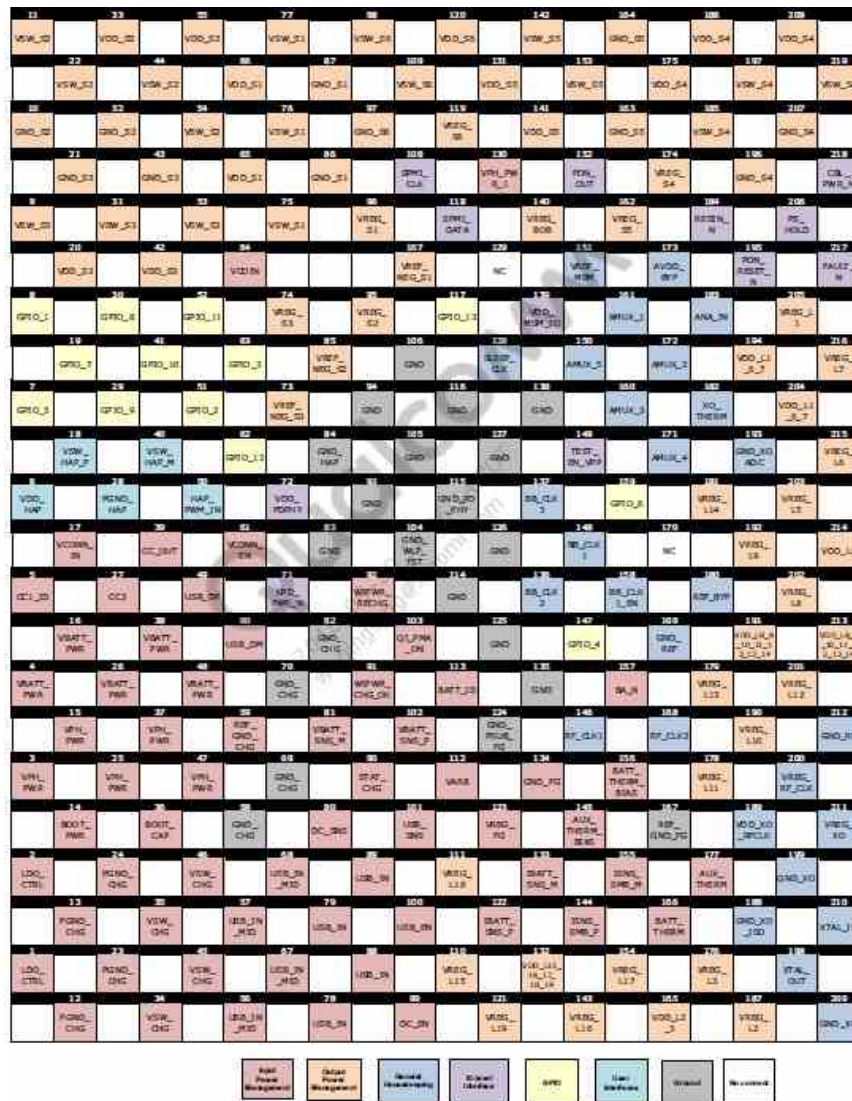


Figure 2-1 PM660 pin assignments (top view)

U1601 chip bitmap:

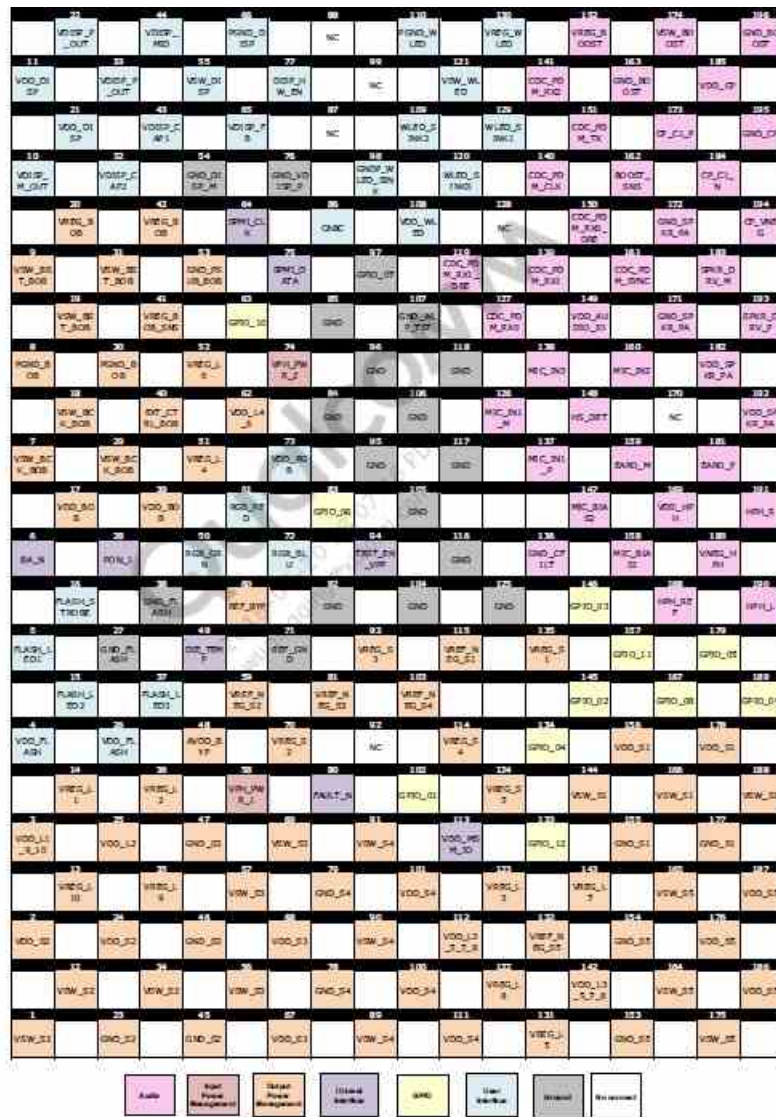


Figure 2-2 PM660L pin assignments (top view)

3. Troubleshooting

3.1 Switching failure

In the process of maintenance without starting up, we should follow the principle of software before hardware, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc. In specific measurement, we should measure according to the starting time sequence.

3.1.1 Constant current without startup

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. If the brushing machine reports an error, measure whether the working conditions of U2001 and U501 are normal (in actual maintenance, the brushing machine 11S reports an error "ACK count don't match", which is generally U2001 damage).
3. If the current is still constant after brushing, measure whether the startup timing and CPU working conditions are normal.

Maintenance Case 1

Failure phenomenon: Do not start up,
Baimiding Android failure cause:
software upgrade

Maintenance method: The Android interface of mobile phone startup is to start the mobile phone software and detect the hardware. First, the machine should be brushed to eliminate software faults.

Maintenance Case 2

Fault Phenomenon:
No Startup Failure
Cause: U2001

Maintenance mode: start up 70mA-150mA constant current, brush the machine for 12S to report error, measure the normal working voltage of CPU, judge the damage of EMMC body, replace and repair the fault.

Maintenance Case 3

Fault Phenomenon:
White Rice Fixed
Screen Fault Cause:
U2001

Maintenance mode: Start-up white rice screen fixing, identify that the port brushes normally, start-up white rice screen fixing software upgrade is invalid, welding CPU and EMMC are invalid, and it is judged that EMMC damage causes the mobile phone software to fail to start and replace U2001 fault repair.

Maintenance Case 4

Failure Phenomenon:
Recovery Mode Failure
Cause: Software Upgrade

Maintenance mode: Start up and enter Recovery mode to brush the machine for repair.

3.1.2 No startup current is not maintained

Analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether the startup timing signal is normal.
3. Measure whether the power supply of U2001 is normal.
4. Measure whether the USB signal circuit is normal (BUS_USB_SNS; USB_HS_D_P_CP; USB_HS_D_M_CP; Note: If these measured signals are abnormal, CPU fault can be judged from the side).

Maintenance Case 1

Failure

Phenomenon: No

Startup Failure

Cause: C1805

Maintenance mode: 100mA startup is not maintained, the brush machine does not recognize the port, measure the startup timing voltage VREG_S4A_2P04 short circuit to ground, remove the circuit C1805 fault repair.

3.1.3 No startup, no current

Analysis ideas:

1. Check whether the appearance of J2803 is damaged, and if the interface is normal, measure whether the ground value of J2803 is normal (BAT_CON_ID, PCB_BATT_THERM, VBAT).
2. Power-on measurement VPH_PWR output is normal, KYPDPWR_N (TP2805) 1.8 V voltage is normal. If there is no output, replace U1201.
3. Measure whether the startup timing signal is normal.

Maintenance Case 1

Failure

Phenomenon: No

Startup Failure

Cause: C1305

Maintenance mode: No current when starting up (the accuracy of power meter is limited, and the actual phenomenon shall prevail), measure the short circuit of VPH_PWR to ground, and remove C1305 for repair.

Maintenance Case 2

Failure

Phenomenon: No

Startup Failure

Cause: C2833

Maintenance mode: No current when starting up (the accuracy of power meter is limited, and the actual phenomenon shall prevail), measure the short circuit of VBAT to ground, and remove C2833 for repair.

3.1.4 Leakage without starting up

Maintenance analysis ideas:

1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.
2. Measure whether VBATT and VPH_PWR are short-circuited. If both VBATT and VPH_PWR are short-circuited, first find VBATT short-circuited elements, then find VPH_PWR short-circuited elements.
3. Measure whether there is short circuit in other power supply lines, and find out the faulty components according to the short circuit signal.
4. Power up to find heating elements.

Maintenance Case 1

Fault phenomenon: leakage 500mA

Faulty components: U3003, U3701

Maintenance analysis: 500mA electric leakage, check that U3003 burned, 400mA electric leakage after removal, check that U3701 is hot, and it is normal after removal, and repair the failure of

parts.

3.2 Restart failure

Analysis ideas:

1. Check whether the 1.8 V voltage of KYPD_PWR_N signal is pulled low.

2. Software upgrade, troubleshooting software.
3. Use QDUTT tool to detect whether DDR is abnormal.
4. Measure whether I2C to ground value and voltage are normal.
5. Measure whether the clock is normal when U501 is powered.
6. Measure whether the motherboard has abnormal power supply caused by short circuit.
7. Replace U501.

3.3 Crash failure

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Use QDUTT tool to detect whether DDR is abnormal.
3. Measure whether BAT_CON_ID and PCB_BATT_THERM are normal.
4. Measure whether the startup timing is normal.
5. Measure whether the power supply of U501 is normal.
6. Measure whether there is abnormal power supply caused by short circuit on the motherboard.
7. Replace U501.

3.4 Signal fault

Bit Chart of Xiaomi Tablet 4 RF Chip U3401:

	17 GND		32 PRX_MB3		48 PRX_HB1		63 PRX_HB3		80 PRX_LNB		96 VDDA_IP2_RX		110 DRX_UHB		126 DRX_HB2		141 DRX_HB1		156 DRX_MB3		172 GND	
8 GND		25 PRX_MB2		38 PRX_MB4		54 PRX_HB2		71 PRX_HB4		88 PRX_LTEU_UHB		101 DRX_LTEU_UHB		117 DRX_HB4		133 DRX_HB2		148 DRX_MB4		164 DRX_MB2		180 GND
	16 PRX_MB1		31 GND		45 GND		62 GND		79 GND		94 GND		109 GND		125 GND		140 GND		155 GND		171 DRX_MB1	
7 PRX_LB1		24 GND			53 VDDA_IP0_RX2_1		70 VDDA_IP2_RX3						116 GND				147 GND		163 GND		179 DRX_LB1	
	15 PRX_LB2			44 GND		61 GND		78 GND		93 GND		108 VDDA_IP2_RX1		124 GND						170 DRX_LB2		
6 PRX_LB3		23 GND		52 GND		69 VDDA_IP0_RX2						115 GND		132 VDDA_IP0_RX0_1				162 GND		178 DRX_LB3		
	14 PRX_LB4		30 GND	43 VDDA_IP0_RX2		60 TX_FBRX_P		77 VDDA_IP0_RX1		92 GND		107 VDDA_IP0_RX0		123 VDDA_IP2_RX0_1		139 GND		154 VDDA_IP0_RX2		169 DRX_LB4		
5 PRX_LB5		22 GND		37 VDDA_IP0_RX3		51 GND		68 TX_FBRX_M		85 VDDA_IP2_FBRX		100 GND		114 XD_IN		131 GND		146 VDDA_IP0_GND3		161 GND3_IN		177 DRX_LB5
	13 VDDA_IP0_TX0_1		42 GND		59 VDDA_IP0_FBRX		76 GND		91 GND		106 GND		122 VDDA_IP0_ANA2		138 GND		153 GND					
4 GND		21 GND		36 GND		50 VDDA_IP0_FBRX		67 GND		84 GND		99 GND			130 VDDA_IP2_RX0		145 GND		160 GND		176 TX_CH1_LTEU	
	12 GND		29 VDDA_IP0_TX0		41 GND		58 GND		75 VDDA_IP2_ANA5		90 VDDA_IP2_ANA3		105 GND		121 VDDA_IP0_ANA2		137 VDDA_IP0_RX0		152 VDDA_IP0_TX1		168 TX_CH1_UHB	
3 VDDA_IP0_TX0		20 GND		35 VDDA_IP0_ANA1		49 GND		66 GND		83 WIMIS_R_ESETN		98 GND		113 ETDAC_CH1_M		129 GND		144 VDDA_IP0_TX1		159 TX_CH1_MB		175 GND
	11 VDDA_IP0_TX0		26 GND		40 GND		57 W_GRFC_5		74 W_GRFC_2		89 GND		104 ETDAC_CH1_P		120 W_GRFC_5		136 W_GRFC_5		151 GND		167 TX_CH1_HB	
2 TX_CH1_LB1		19 GND		34 GND		48 GND		65 W_GRFC_1		82 VDDA_IP0		97 GND		112 VDDA_IP0		128 GND		143 GLINK_EN		158 GND		174 GND
	10 TX_CH1_MB2		27 TX_CH1_LB2			56 GND		73 GLINK_REQ		88 GLINK_CLK_M		103 GLINK_CLK_F		119 GLINK_CLK_F		135 GLINK_CLK_M		150 GND5_ELNA_CTRL		166 RFFE1_CLK		
1 GND		18 TX_CH1_LNB		33 TX_CH1_MB1		47 ETDAC_CH0_M		64 VDDA_IP0		81 GLINK_CLK_M		96 GLINK_CLK_F		111 GLINK_CLK_M		127 GLINK_CLK_F		142 VDDA_IP0		157 W_GRFC_2		173 GND
	9 GND		25 TX_CH1_HB1		39 ETDAC_CH0_F		55 GND		72 GLINK_CLK_F		87 GLINK_CLK_M		102 GND		118 VDDA_IP0_CLK		134 GND		149 RFFE1_DATA		165 GND	

Figure 2-2 SDR660 pin assignments – top view

Analysis ideas:

1. Insert SIM card to ensure normal identification, and eliminate faults that do not recognize SIM card.

2. Software upgrade, troubleshooting software.

3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.

4. Measure whether the power supply of RF circuit is normal and whether the RFFE CLK (1/2)/DAT1 signal is normal.

5. If the RF calibration is normal and there is still no signal, check the SIM card circuit and whether the J3802 to J3803 lines are normal.

Maintenance Case 1

Fault phenomenon: no signal

Cause of failure: U501, U3401

Maintenance analysis: Check and identify SIM card is normal and has no signal. RF calibration offline XO-cal, after replacing U501, calibrate all items of B34 and B39, and the fault point lags behind. Measure that the power supply of U3401 is normal, and the relevant components of B34 and B39 are invalid. Replace U3401 for fault repair.

Maintenance Case 2

Fault Phenomenon:

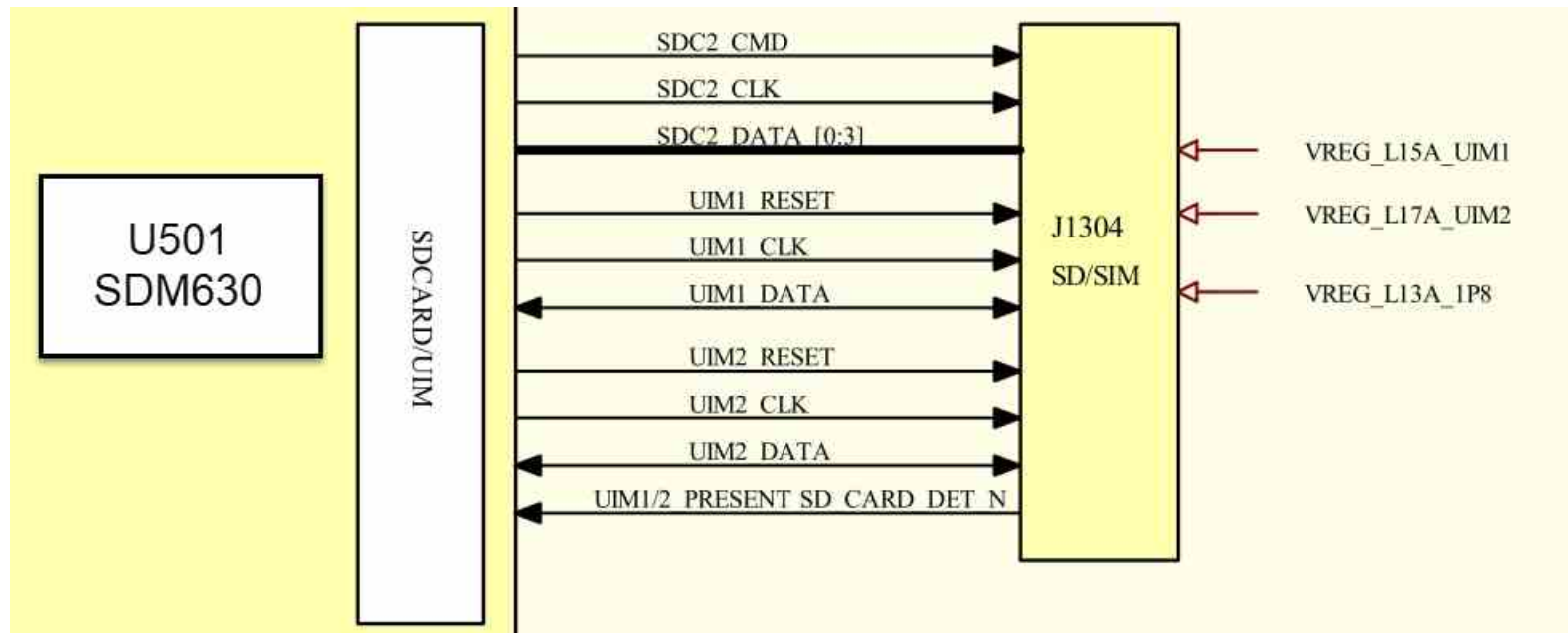
Signal Weak Fault

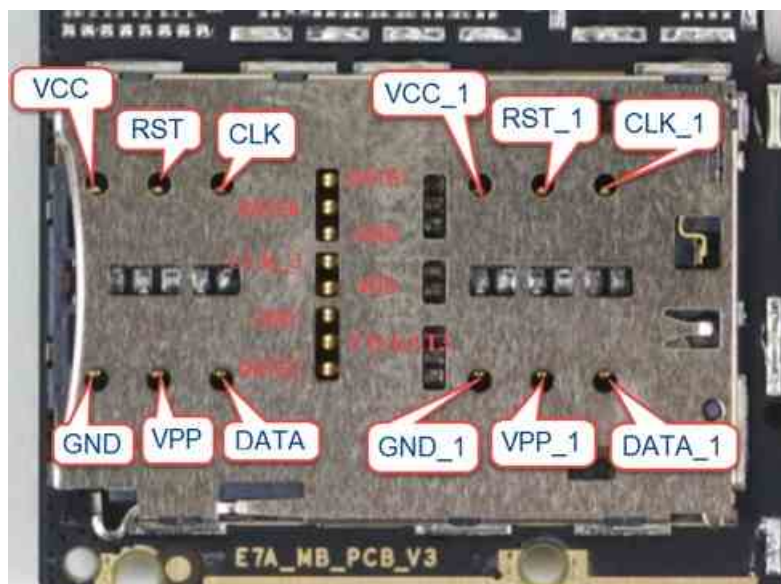
Component: U3003

Maintenance analysis: The detection signal is weak, unstable sometimes, RF calibration LTE-B1, LTE-B3 transmission off-line, replace U3003 fault repair.

3.5 SIM card failure

Principle block diagram of SIM card:





SIM/SD card meter:

SIM/SD测量表		
Symbol	测量值	测量点
VCC_1	650	SIM1
RST_1	430	
CLC_1	430	
VPP_1	1	
DATA_1	430	
VCC_2	650	SIM2
RST_2	430	
CLC_2	430	
VPP_2	1	
DATA_2	430	
DATE1	420	SD
DATE0	420	
CLK_3	439	
VDD	680	
CMD	420	
CD/DAT3	420	
DATE2	420	

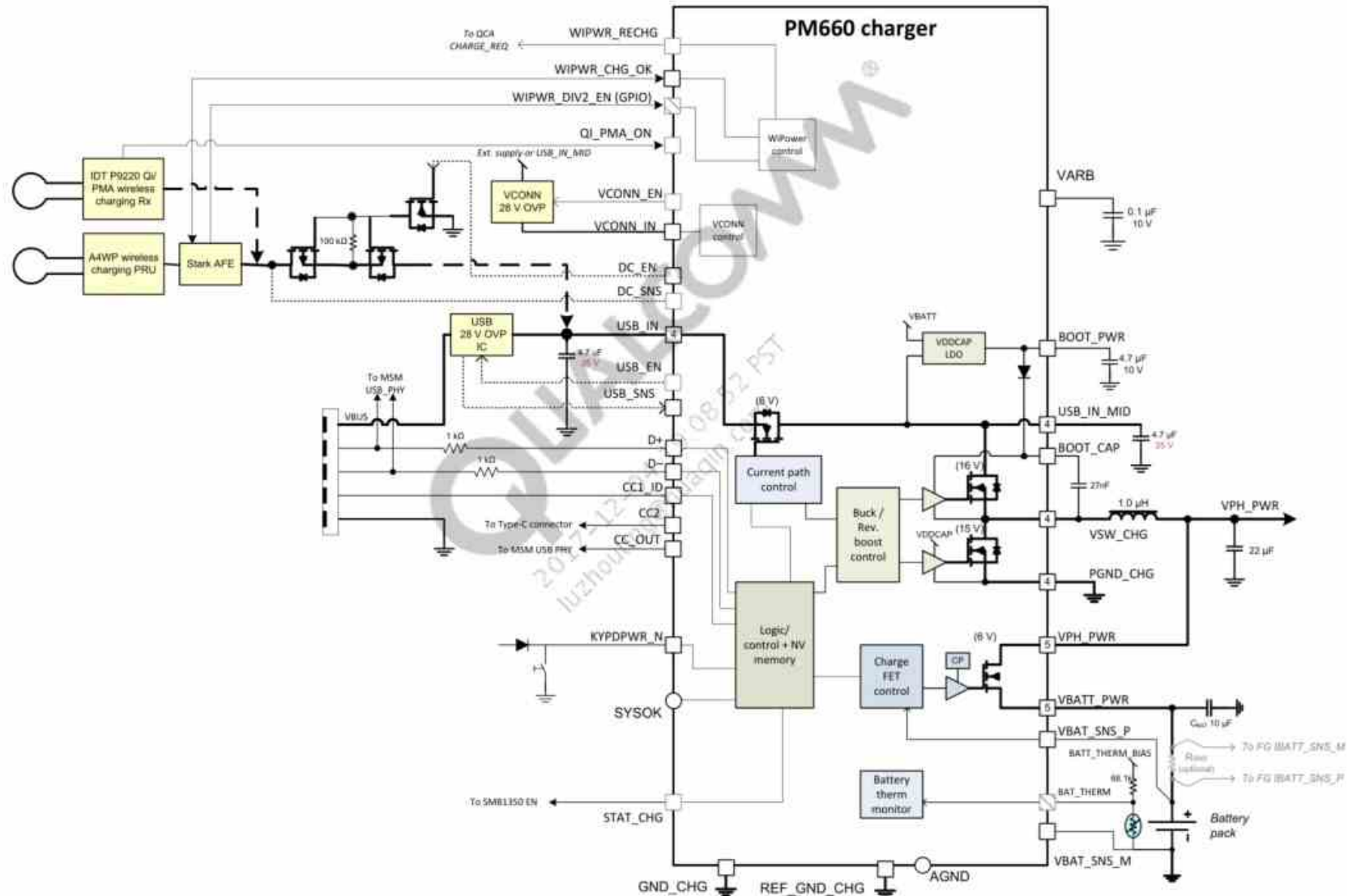
- 1.First, check whether the baseband version of the mobile phone is normal. If the baseband information is normal, it is the SIM card related function failure.
- 2.Check whether the SIM card pin is deformed, oxidized or broken, and carefully observe whether there is virtual welding phenomenon in the solder joint of SIM card holder.
- 3.Measure whether the SIM card aims at the ground value normally.

4. Start up and test whether the power supply, clock and reset of SIM card are normal, and whether the data voltage jump is normal
5. If the above signal is normal, replace U501.

3.6 Charging failure

Charging schematic diagram:

PM660 Charger Block Diagram

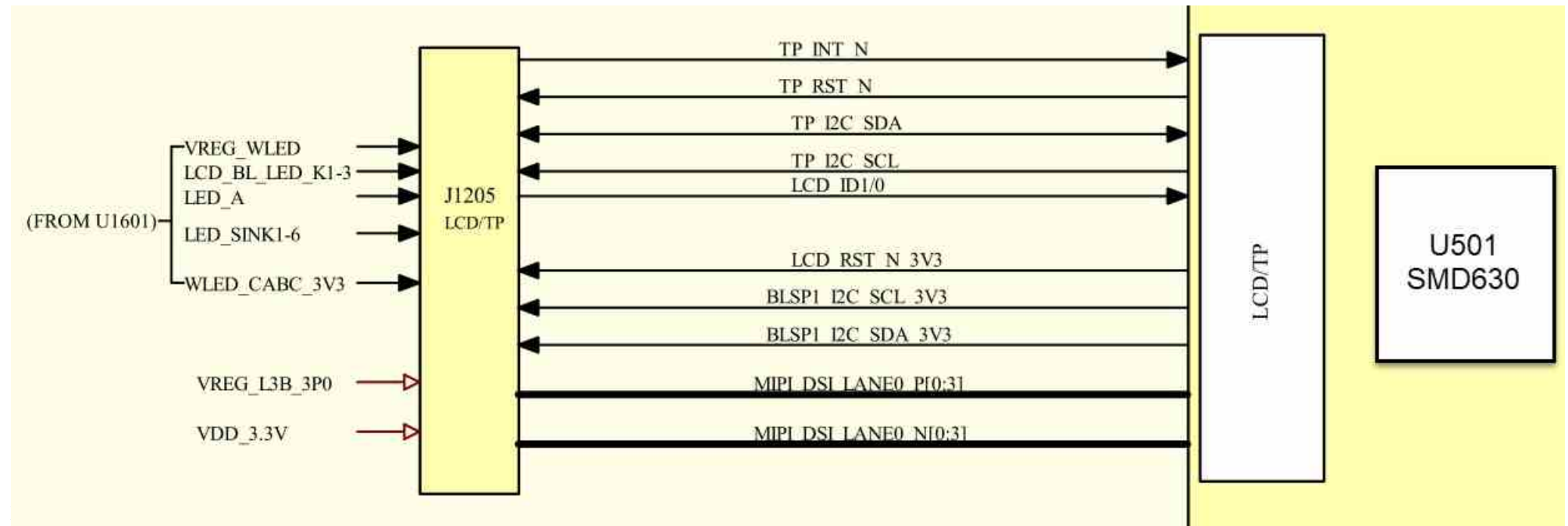


Analysis ideas:

1. Check whether the appearance of J2802 and J2803 is normal, and measure whether the ground values of VBUS_USB_SNS, USB_HS_D_P_CON and USB_HS_D_M_CON are normal with multimeter diode gear.
2. Measure whether VBAT, AT_CON_ID and PCB_BATT_THERM of J2803 are normal to ground.
3. To ensure that the above signals are normal, consider replacing U1201.

3.7 Display failure and touch screen failure

Principle block diagram:



GND	1	550	550	550	1	1	GND	410	410	410	410	1	532	532	532	532	1	596	596	596	1	1	1	GND
25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
J2502																								
26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
GND	260	260	GND	260	260	GND	260	260	GND	260	260	GND	260	260	GND	1	1100	990	990	1	600	600	600	GND

Display some measurement tables:

显示电压测量表		
Symbol	测量值	测量点
LDM_ID1	4V-29V	R2521
WLED_CABC_3V3	3.3V	R2535
LCD_RST_N_3V3	3.3V	R2510
VDD_3.3V	3.3V	C2527
BLSP1_I2C_SDA_3V3	3.3V	J2502第47脚
BLSP1_I2C_SCL_3V3	3.3V	J2502第48脚
LCD_BL_LED_K1	0.3V	R2501
LCD_BL_LED_K2	0.3V	R2502
LCD_BL_LED_K3	0.24V	R2517

voltage measurement of touch screen part:

TOUCH电压测量表		
Symbol	测量值	测量点
TP_I2C_SDA	1.8V	R808
TP_I2C_SCL	1.8V	R809
TP_RST_N	1.8V	R2514
TP_INT_N	1.8V	R2513
VREG_L3B_3P	3.3V	R2509

Maintenance analysis ideas:

1. Visually check whether J2502 and its peripheral components are damaged or falsely welded.
2. Brush the machine to eliminate software faults.
3. Measure whether the ground value of each pin of J2502 is normal with multimeter diode gear.
4. If the ground value is normal, measure whether the power supply and control signals in the "Display Measurement Table" are normal. If the ground value is normal,
5. Replace U501.

Maintenance Case 1

Fault Phenomenon:

Black Screen No

Display Fault

Component: C1903

Maintenance analysis ideas: There is no display on the black screen of the motherboard, and there is no high voltage output on the measurement display boost VREG_WLED. Replace C1903 for fault repair, and replace the display boost circuits L1901 and D1901 at the same time as required.

Maintenance Case 2

Fault Phenomenon:

Black Screen No

Display Fault

Component: U1601

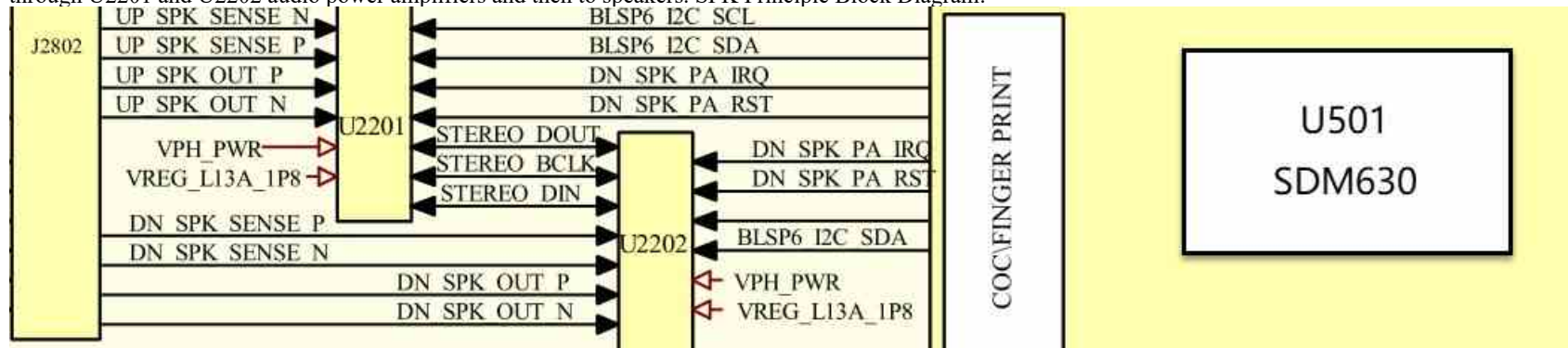
Maintenance analysis ideas: there is no display on the black screen of the motherboard, and there is no high voltage output on the booster VREG_WLED according to measurement. It is invalid to replace C1903, L1901 and D1901, and it is invalid to replace U1601 for fault repair.

3.8 Audio fault

The audio circuit of Xiaomi Tablet 4 includes: dual speakers, microphones and earphones (excluding earphones). After the audio signals come out of U1601, they go to different channels respectively. First, distinguish which part has the problem according to the fault phenomenon, and then analyze and repair according to the following modules.

3.8.1 Loudspeaker failure

Speaker is connected to the motherboard through FPC. Its principle is to go to CODEC through CPU first, and then amplify and output to interface J2802 through U2201 and U2202 audio power amplifiers and then to speakers. SPK Principle Block Diagram:

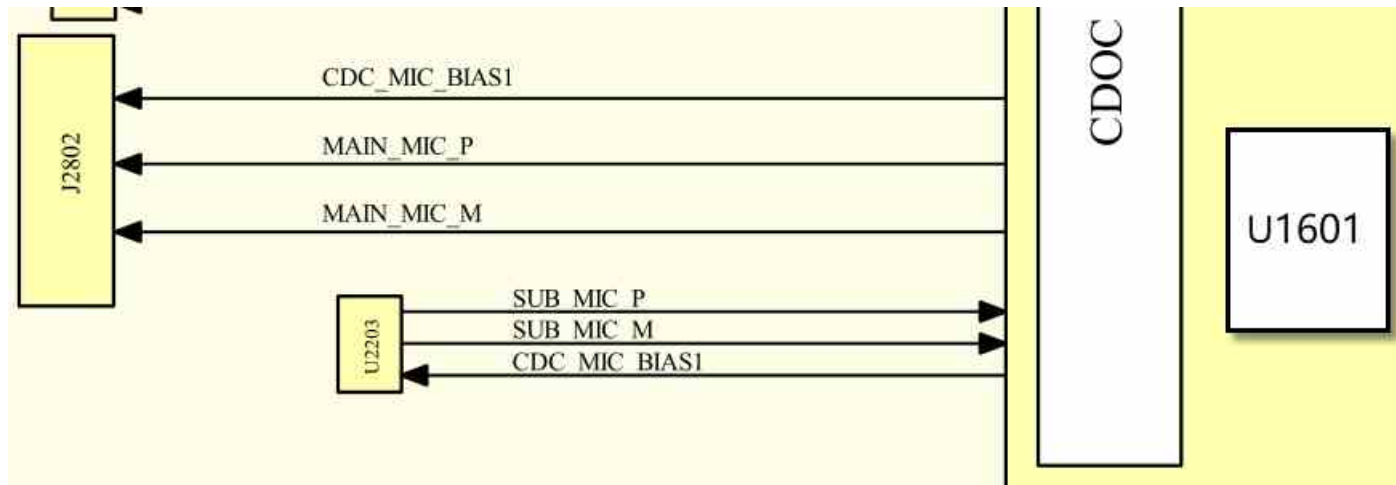


Maintenance analysis ideas:

- 1 Visually check whether J2802 looks good.
2. Software upgrade troubleshooting software.
3. Make sure that one of the two speakers is faulty.
4. Measure whether the working conditions of U2201 and U2202 are normal.
4. If the above signals are normal, consider whether the bus from CODEC to CPU is normal.

3.8.2 MIC fault

The main board of Xiaomi Tablet 4 includes two MIC circuits, main/auxiliary MIC, and the main MIC is welded on the auxiliary board. MIC meter:



Maintenance analysis ideas:

1. Visually check whether J2802 looks good.
2. Measure whether MAIN_MIC_P, MAIN_MIC_M and CDC_MIC_BIAS1 are normal to ground.
3. Measure whether the CDC_MIC_BIAS1 voltage is normal.
4. If the above signals are normal, consider whether the bus between CODEC and CPU is normal.

3.8.3 Headphone failure

Maintenance analysis ideas:

1. Check whether the earphone interface contacts are deformed and whether there are foreign bodies in the earphone interface.
2. Software upgrade and troubleshooting.
3. Measure whether the earphone interface is normal to ground, do not insert the earphone when starting up, and measure whether the DETECT detection signal has 1.8 V voltage.
4. Repair the headphone channel according to the principle block diagram, and replace U1601 normally.
5. Replace U1601.

Schematic diagram of earphone interface:



Maintenance Case 1

Fault Phenomenon:

Message Noise Fault

Component: U2203

Maintenance analysis ideas: Self-checking MIC function sometimes can not record, occasionally can record but the sound is small and noisy, replace noise reduction MIC U2203 fault repair.

3.9 WIFI/BT/FM/GPS Fault

U4001 Bitmap:

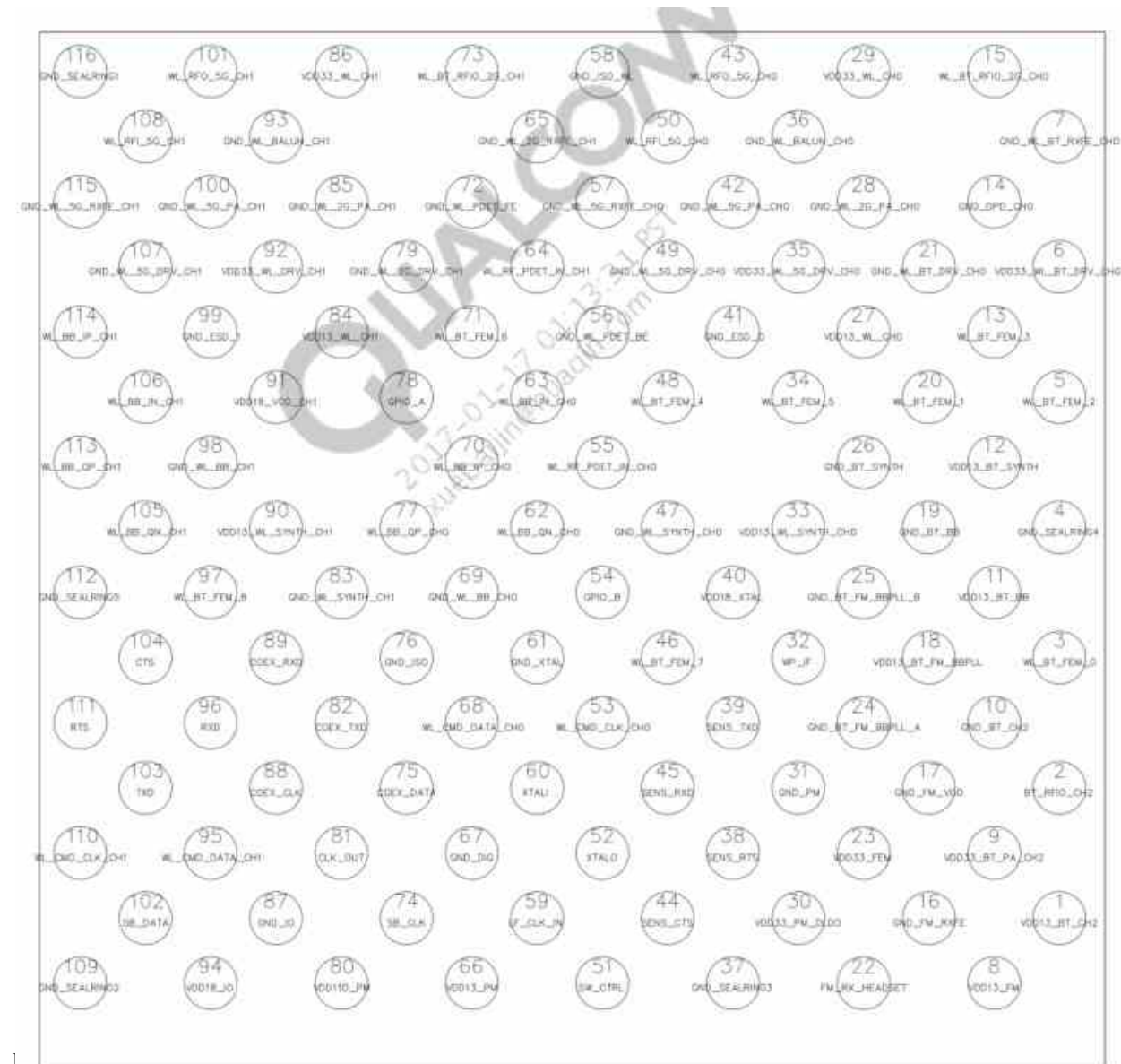


Figure 2-1 WCN3990 ball assignments (bottom view)

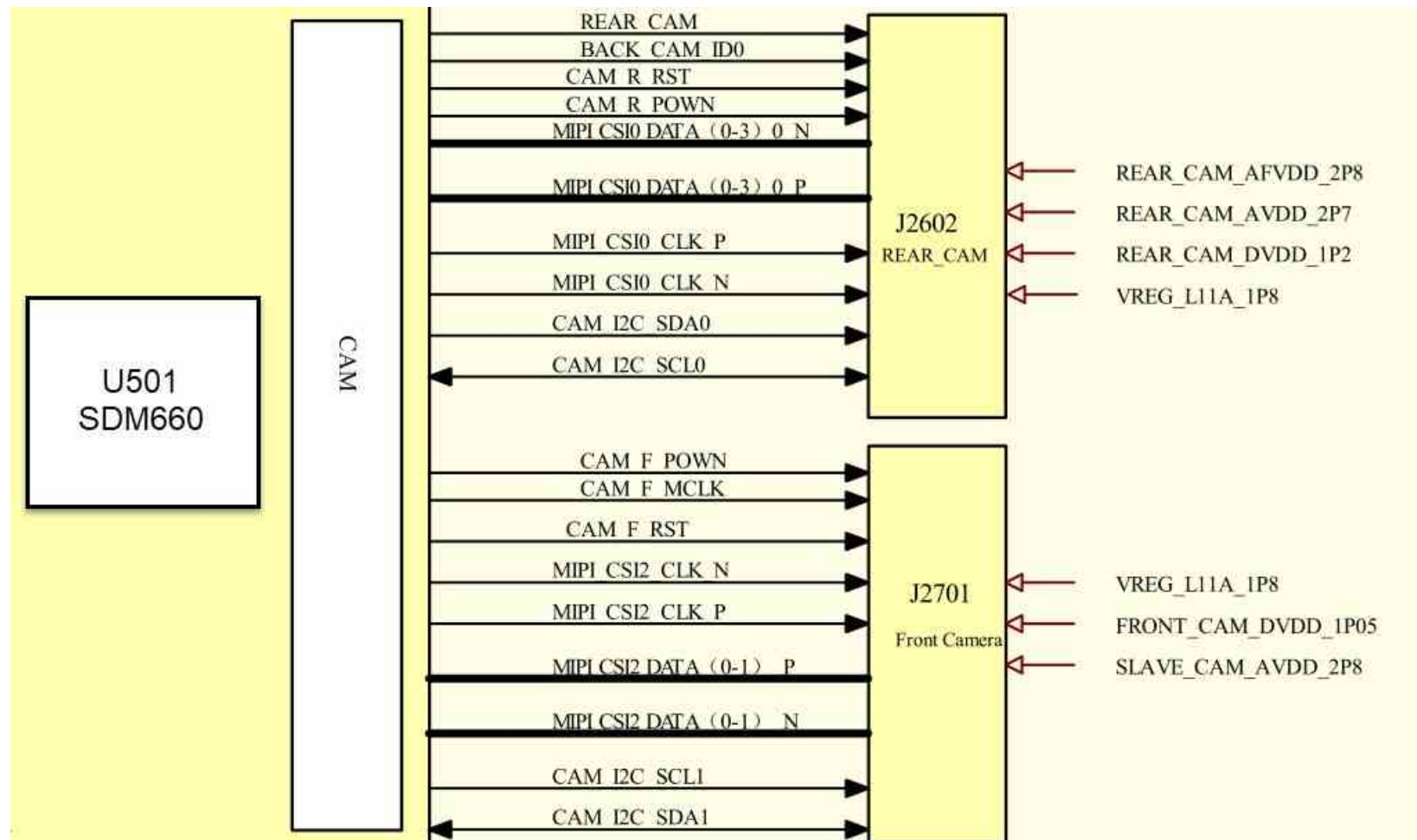
U4001电压测量表		
Symbol	测量值	测量点
VREG_L6A_1P3	1.3V	R4016
REG_L13A_1P8	1.8V	C4015
VREG_L9A_1P8	1.2V	C4012
VDD11D_PM	1V	C4005
VREG_L19A_3P3	3.3V	R4015
VREG_L8B_3P3	3.3V	C4032
SLEEP_CLK	32KHz	C3007

Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether X4001 is normal (48MHz).
3. Measure whether the power supply, clock and enable signals of U4000 are normal.
4. Take off U4001 and measure whether the bus between U2001 is normal. If it is normal, replace U4001.
5. Replace U501.

3.10 Camera failure

Principle block diagram:



Front camera signal measuring table:

前置相机电压测量表		
Symbol	测量值	测量点
SLAVE_CAM_AVDD_2P8	2.8V	C2705
FRONT_CAM_DVDD_1P05	1.05V	C2707
CAM_I2C_SCL1	1.8V	R807
CAM_I2C_SDA1	1.8V	R806
CAM_F_RST	1.8V	C2702
CAM_F_MCLK	1V	R2701
MIPI_CSI2_CLK_P/N	760	Z2705
MIPI_CSI2_DATA0-1 P/N	760	Z2703/Z2704

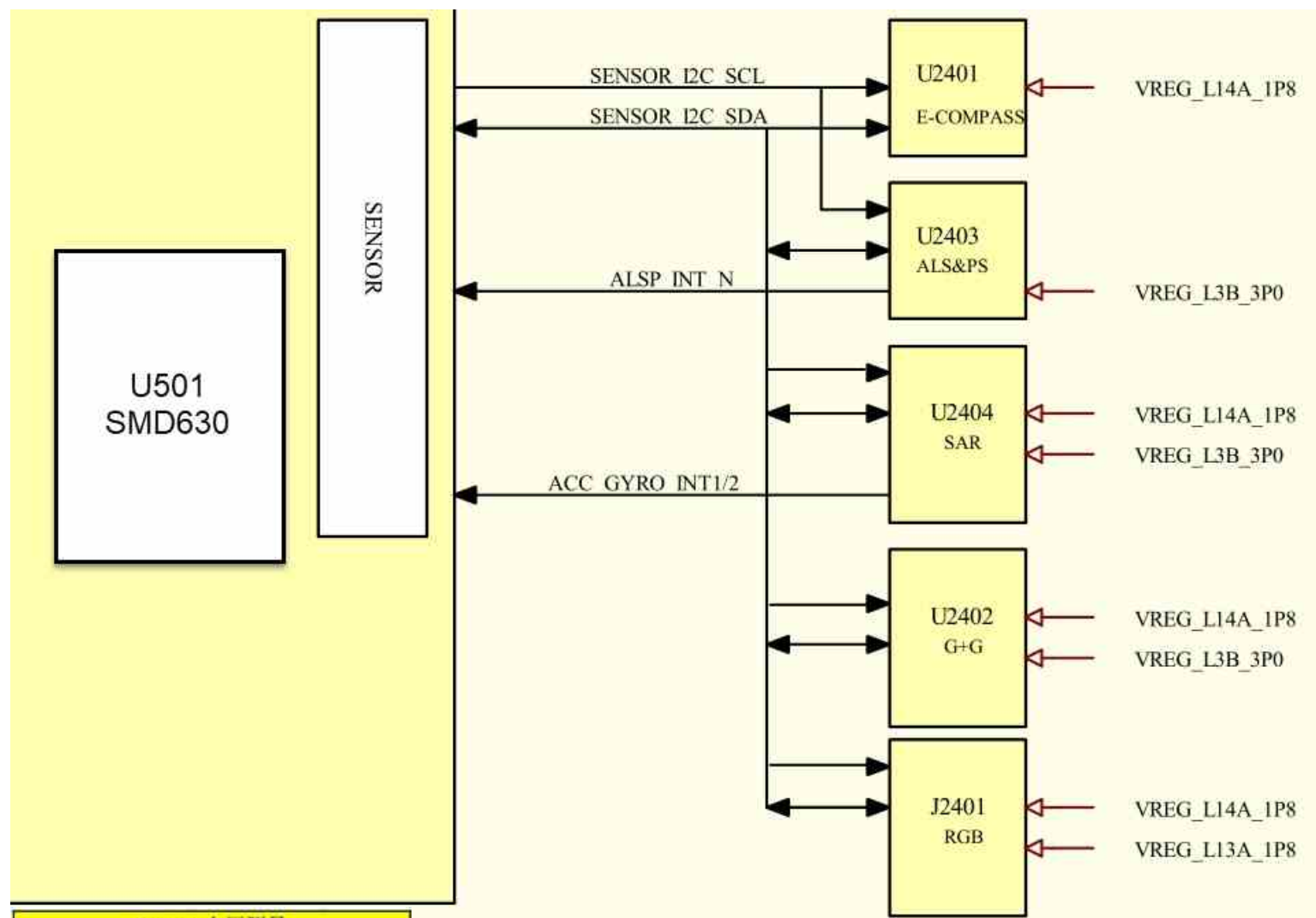
Rear camera signal measurement:

后置相机电压测量表		
Symbol	测量值	测量点
REAR_CAM_AFVDD_2P8	2.8V	C2712
REAR_CAM_AVDD_2P7	2.7V	R2608
VREG_L11A_1P8	1.8V	R2610
REAR_CAM_DVDD_1P2	1.2V	C2614
CAM_I2C_SDA0	1.8V	R805
CAM_I2C_SCL0	1.8V	R804
CAM_R_RST	1.8V	C2605
CAM_R_MCLK	1V	R2602
MIPI_CSI0_DATA0-3_N/P	760	Z260 (2/4/6/8)
MIPI_CSI0_CLK_P/N	760	Z2610

1. Software upgrade, troubleshooting software.
2. Detect J2602, J2701 and surrounding components for loss and damage.
3. Go to CIT to test the front camera and the rear camera to distinguish faults.
4. Measure whether the ground values of J2602 and J2701 are normal.
5. Measure whether the camera power supply, clock and reset signal output are normal.
6. Measure whether the I2C and MIPI bus output by U501 are normal.

3.11 Inductor failure

Principle block diagram:



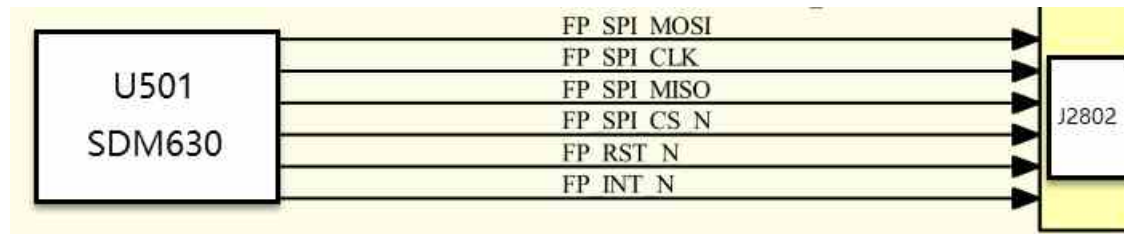
Sensor电压测量		
Symbol	测量值	测量点
SENSOR_I2C_SCL	1.8V	R810
SENSOR_I2C_SDA	1.8V	R811
VREG_L14A_1P8	1.8V	C1416
VREG_L3B_3P0	3V	C1710
VREG_L13A_1P8	1.8V	C2590

Maintenance analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether the working conditions of the corresponding sensors are normal.
3. Replace the corresponding sensor.
4. Replace U501.

3.12 Fingerprint identification failure

Principle block diagram:



Fingerprint voltage meter:

指纹电压测量表		
Symbol	测量值	测量点
FP_VDDIO	1.8V	C2805
FP_AVDD	2.8V	C2806
FP_SPI_MISO	1.8V	J1900第23PIN
FP_RST_N	1.8V	C2807

fingerprint interface and surrounding components are damaged, and if there is damage, please replace it.

2. Brush the machine to eliminate software faults.
3. Measure whether the above voltage and other signals are normal.
4. If all the above signals are normal, replace U501.